

Predictive Model for Seaweed Biomass Harvesting to Mitigate Environmental Carbon Dioxide Levels.

1. PROBLEM STATEMENT-

We aim to develop a Predictive Model to optimize seaweed harvesting for carbon sequestration. We can identify optimal harvesting times and practices by analyzing seaweed biomass data and environmental factors. This will enhance carbon capture efficiency and protect marine biodiversity. Our work is relevant to Climate Science, Biodiversity, and Agriculture.

2. RESEARCH AND LITERATURE REVIEW

2.1 Large global variations in the carbon dioxide removal potential of seaweed farming due to biophysical constraints by Isabella B. Arzeno- Soltero, B. Saenz, C. Frieder, M. Long, J. DeAngelo, S. Davis, K. Davis. (1)

AIM: The aim of this study is to estimate the global potential yield and carbon dioxide removal potential of seaweed farming.

DATA: The data in this paper is the estimates and modeling of the global potential yield of seaweed farming for carbon dioxide removal.

METHOD: Used a dynamic seaweed growth model that incorporates growth-limiting factors such as nitrate supply - Estimated the global potential yield of four different seaweed types using the growth model - Calculated the required cultivation area to achieve specific levels of seaweed carbon harvesting, and found that efficiency decreases as the area expands beyond the most productive regions - Identified biophysical constraints, such as seaweed loss rates due to infestation, disease, grazing, and wave erosion, as an area requiring further investigation to improve the accuracy of seaweed yield estimates

INNOVATION: The key innovation in this paper is the use of a dynamic seaweed growth model that includes growth limiting mechanisms, such as nitrate supply, to estimate the global potential yield of four types of seaweed.

RESULT: - Harvesting 1 Gt of seaweed carbon per year would require over 1 million km² of the most productive ocean areas, and tripling the cultivation area would only yield an

additional 1 Gt of carbon, indicating diminishing returns. - Accurately estimating the annual seaweed harvest yield requires a better understanding of the biophysical constraints that limit seaweed growth and survival, such as infestation, disease, grazing, and wave erosion.

EVALUATION: The study by Arzeno- Soltero et al. highlights significant regional variability in the carbon removal potential of seaweed farming due to factors like water temperature, nutrients, and ocean currents. Using global models, it shows that biophysical constraints limit scalability in many areas. The research emphasizes the need for region-specific strategies to optimize seaweed farming for effective climate mitigation.

AIM: To discuss the concept of coastal marine biorefineries and their merits over conventional biorefinery systems. 2. To analyze the potential of seaweed for CO₂ sequestration by calculating the time required to remove 100 GT of atmospheric CO₂ through seaweed farming. The paper also aims to determine the total bioethanol and high value chemicals (HVC) that may be produced from the produced seaweed biomass.

DATA: The key data in this paper consists of information and values related to seaweed biomass, carbon sequestration, and the production of biofuels and high value chemicals from seaweed, including seaweed net primary productivity, carbon to biomass conversion factors, bioethanol conversion factors, and market prices for various high value compounds

METHOD: The key methods used in this study were: 1) Literature review to identify relevant technologies for a coastal marine biorefinery, 2) Scenario analysis to assess seaweed's carbon sequestration potential under different cultivation area assumptions, and 3) Calculations using equations and literature-derived inputs to estimate time, biomass, carbon removal, and product yields.

INNOVATION: The key innovation described in this paper is the concept of a "coastal marine biorefinery" that utilizes marine resources such as seawater, seaweed, and marine microorganisms to produce biofuels and high-value chemicals in a more sustainable and cost effective manner compared to conventional biorefineries.

RESULT: Seaweeds can be used to efficiently remove 100 Gt of atmospheric CO₂ within 12 years or less, depending on the seaweed species and cultivation area used. - Seaweed cultivation on ecologically available ocean areas (48 million km²) could remove 100 Gt of CO₂ in less than a year. - The biomass from seaweed cultivation could produce sufficient bioethanol to meet global energy demands and generate multi-billion-dollar revenue from high-value chemicals.

EVALUATION: The study "Recent Advances in Seaweed Biorefineries and Assessment of Their Potential for Carbon Capture and Storage" highlights seaweed farming as a promising method for carbon sequestration and biofuel production. Coastal marine biorefineries can utilize seaweed to

capture up to 100 gigatons of CO₂, produce bioethanol (up to 8 trillion liters), and generate valuable chemicals, creating opportunities for sustainable economic growth and climate change mitigation

2.2 Recent Advances in Seaweed Biorefineries and Assessment of Their Potential for Carbon Capture and Storage Katherine G. Johnston, Abdelfata h Abomohra, Christopher E. French, A. Zaky (2)

2.3 Economic and biophysical limits to seaweed farming for climate change mitigation J. DeAngelo , Benjamin T. Saenz, Isabella B. Arzeno- Soltero, C. A. Frieder, M. Long, Joseph Hamman, Kristen A. Davis, Steve Davis (3)

AIM: To assess the biophysical and economic limits to farming seaweed at scales relevant to the global carbon budget.

DATA: The "data" in this paper consists of the outputs and results from running coupled seaweed growth and technoeconomic models, including estimates of the costs of seaweed farming for carbon sequestration and emissions avoidance, as well as the estimated area of seaweed farming required to achieve gigaton scale climate benefits.

METHOD: The method used in this study was to use coupled seaweed growth and technoeconomic models to estimate the costs and climate benefits of global seaweed production, systematically testing different model parameters and assumptions.

INNOVATION: The key innovation in this paper is the use of coupled seaweed growth and technoeconomic models to estimate the costs and climate benefits of global seaweed production at scales relevant to the global carbon budget.

RESULT: The key result of the paper is an estimate of the costs and climate benefits of large-scale seaweed farming for carbon sequestration and emissions avoidance, with the authors finding that under optimistic assumptions, sinking farmed seaweed could sequester 1 gigaton of CO₂ per year at a cost of \$480 per ton, while using seaweed products to avoid 1 gigaton of CO₂-equivalent emissions could return a profit of \$50 per ton. However, the authors note that these results depend on several critical assumptions and that achieving gigaton-scale climate benefits would require a >30-fold increase in the current area of seaweed farming.

EVALUATION: The study by J. DeAngelo et al. (2022) evaluated the economic and biophysical limits to seaweed farming for climate change mitigation.

AIM: The aim of the paper is to discuss the potential benefits and opportunities of expanding seaweed aquaculture in tropical developing countries.
DATA: Seaweed aquaculture in tropical developing countries enhances food security by providing a rich source of essential nutrients like iodine, calcium, and vitamins, while also serving as a sustainable, low cost food supplement
METHOD: The method described in the paper is a general overview of seaweed farming approaches, including small-scale intertidal farming and larger-scale offshore farming using hatchery produced seedlings. The paper also provides more detailed information on a specific seaweed production system developed by Seaweed Seed Supply A/S, which includes the key steps of spawning, hatchery rearing, grow-out, and harvest. harvesting, and found that efficiency decreases as the area expands beyond the most productive regions - Identified biophysical constraints, such as seaweed loss rates due to infestation, disease, grazing, and wave erosion, as an area requiring further investigation to improve the accuracy of seaweed yield estimates
INNOVATION: The key innovation described in this paper is the development of a modular, floating seaweed farming system that can be deployed offshore in tropical regions, addressing constraints around marine spatial planning, financing, and competition for inshore space.
RESULT: - Seaweed aquaculture in tropical developing countries boosts food security by providing a sustainable, nutrient-rich food source and generating income through local and export markets. It also benefits environmental health by improving water quality, reducing coastal erosion, and absorbing excess nutrients, supporting coastal ecosystem resilience.
EVALUATION: The author highlights seaweed aquaculture as a promising tool for food security, economic growth, and environmental sustainability in tropical developing countries, but stresses the need to address challenges like market access, management practices, and climate change for long-term success.

3. METHODOLOGY:

3.1.1 Factors Affecting Seaweed Growth and Carbon Sequestration-

Water Temperature: Water temperature directly impacts seaweed metabolism. Warmer waters can accelerate growth but may reduce net carbon uptake. Conversely, colder temperatures can slow growth but improve carbon sequestration.

Salinity Levels: Seaweed health is sensitive to salinity fluctuations. Monitoring salinity levels is crucial for optimizing growth and preventing stress.

Climate: Climate conditions, including temperature, precipitation, and wind patterns, influence seaweed growth patterns and harvesting cycles. (10).

Location: Geographic location significantly impacts growth rates and carbon sequestration potential. Factors like water depth, nutrient availability, light intensity, and water currents play a crucial role. (10)

Nutrient Availability: Seaweed growth is dependent on nutrient availability, primarily nitrogen and phosphorus. Monitoring nutrient levels and optimizing fertilization strategies is essential.

Light Exposure: Light is crucial for photosynthesis. Adequate light exposure is essential for maximizing growth and carbon sequestration.

Growth Rate of Different Species: Different seaweed species exhibit varying growth rates. Understanding species-specific growth characteristics is crucial for selecting the most suitable species for cultivation.

Current and Wave Action: Currents and waves influence nutrient supply and physical stress on seaweed. Monitoring these factors can help optimize cultivation practices and improve growth.

Carbon Absorption Rates: Carbon absorption rates vary among different seaweed species and environmental conditions. Understanding these rates is crucial for assessing the potential of seaweed cultivation for carbon sequestration.

3.1.2 Data collection and preprocessing approaches-

Remote Sensing and GIS: Remote sensing techniques, such as those employed by Sentinel-2, MODIS (9) , and Landsat-8 satellites, are instrumental in monitoring large-scale seaweed growth patterns. By analyzing spectral data, particularly using the Normalized Difference Vegetation Index (NDVI) and Algal Bloom Index (ABI), it is possible to extract valuable information on seaweed biomass, chlorophyll concentration, and growth areas.

Environmental Data: Environmental factors significantly influence seaweed growth. Data sourced from the NOAA World Ocean Database (8) and Copernicus Marine Environment Monitoring Service provide crucial insights into key variables like temperature, pH, salinity, nutrient concentrations, and light penetration.

In-Situ Measurements : Complementing remote sensing data, in-situ measurements using CTD sensors and spectroradiometers offer real-time monitoring of seaweed growth, CO₂ uptake rates, and biomass yield. This granular data is essential for optimizing cultivation practices and understanding the ecological impact of seaweed farming.

3.1.3 Feature Selection and Engineering-

Careful feature selection and engineering are crucial to effectively model seaweed growth and productivity. This involves identifying the most relevant factors and transforming raw data into meaningful features.

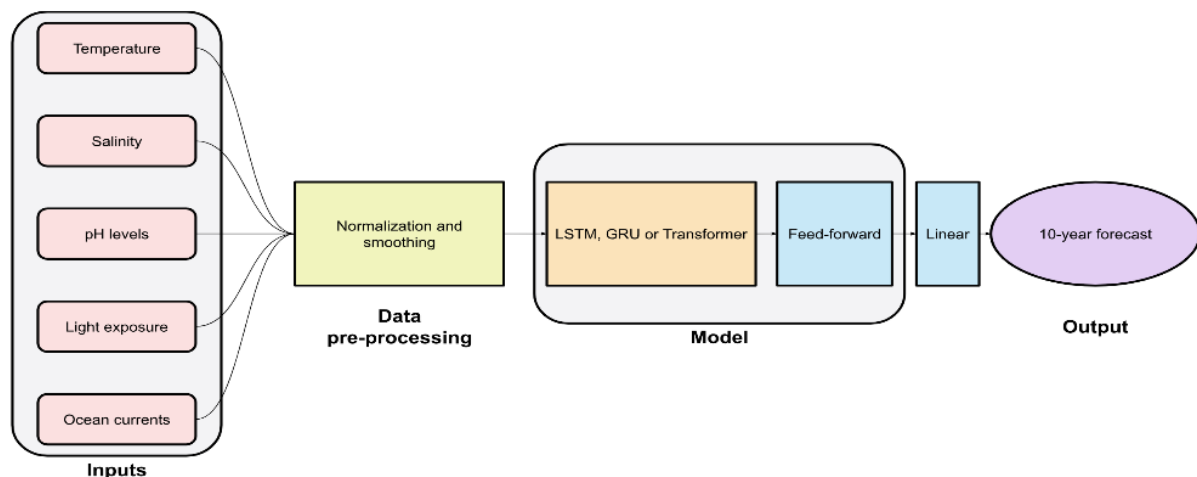
Dimensionality Reduction: Principal Component Analysis (PCA) is a powerful technique used to reduce the dimensionality of the dataset while retaining the most significant information. By identifying the principal components, we can highlight the key factors influencing seaweed growth, such as nutrient availability, light intensity, and water quality.

Preprocessing data: normalizing standard data, creating sliding windows, resizing images, applying algorithms like KNN and mean imputation to fill and fix missing and skewed data.

3.1.4 Predictive modelling-

Model selection - we will be using a mix of model architectures

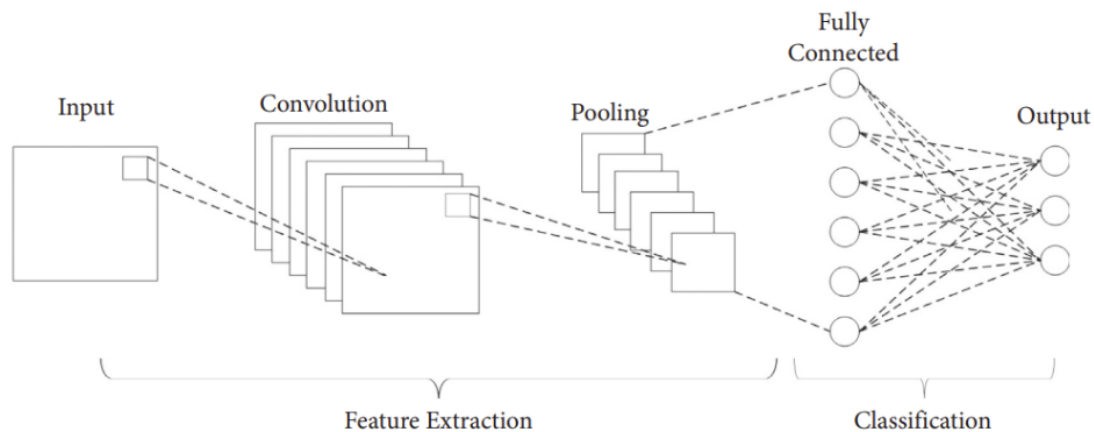
- **Time-Series prediction:** Model choice: LSTM, GRU or Transformer (if enough data). Model purpose: forecasting growth and CO2 sequestration over the 10 year period. Input(s): historical data on environment conditions and current growth rates in given region. Output(s): growth and sequestration over time by the seaweed
- **Spatial analysis:** Model choice: CNN. Model purpose: analyze satellite imagery/geospatial data for finding suitable regions. Input(s): Satellite images of the ocean. Output(s): Suitable regions to start farming



Let's dig deeper into how a CNN-LSTM model would work to predict sequestration

1) **CNN:** For analyzing satellite images and mapping seaweed growth patterns from spatial data. Its main feature is the ability to process multichannel input data, so it is ideal for handling different timeseries data with multiple inputs and outputs(11).

Convolutional Layer: extracts important features from input time series. Kernel size determines the window size for feature extraction. Pooling Layer: reduces dimensionality of feature maps, capturing the most salient. Flatten Layer: convert into 1d vectors so they can be fed into LSTM layer.



2) **LSTM:** For predicting temporal patterns, especially for time-series data like growth rates, seasonal variations and environmental factors. The LSTM algorithm is ideal for classifying, sorting, and making predictions from a single time-series dataset (11). Lstm layers: process the input data sequentially, capturing long term dependencies. number of neurons in the layer would depend on the complexity of data. Dropout layer: to prevent overfitting. Dense layers: to aggregate features learned by LSTM units. Transforms the output into a suitable representation for prediction

Output layer: predicts carbon sequestration potential.

The basic structure of an LSTM model is given below (11).

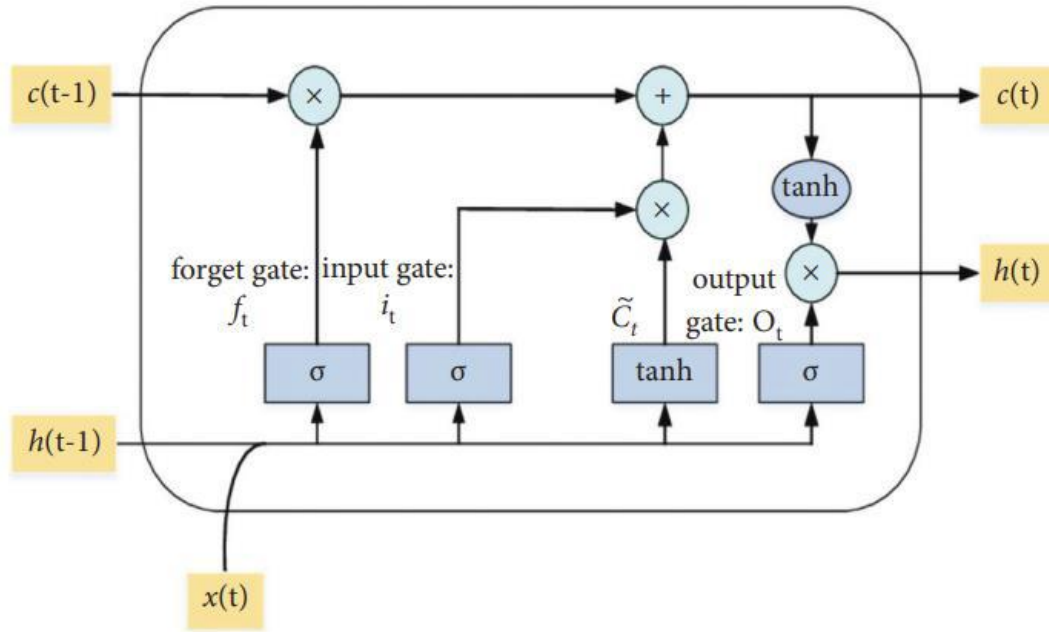


FIGURE 2: Basic architecture of LSTM when being used for time-series prediction in a supervised model. The figure is recreated from

Integrating these models into one

- Using multi-modal architectures to fuse proposed models in-order to combine the numerical, temporal, and geospatial data into a single model. Why? It's simpler during inference.
- Both LSTM and CNN can be used together to create deep learning models that can investigate complicated and unknown patterns in large and varied data sets.

Below is a figure depicting the structural diagram of the CNN-LTSM model (11).

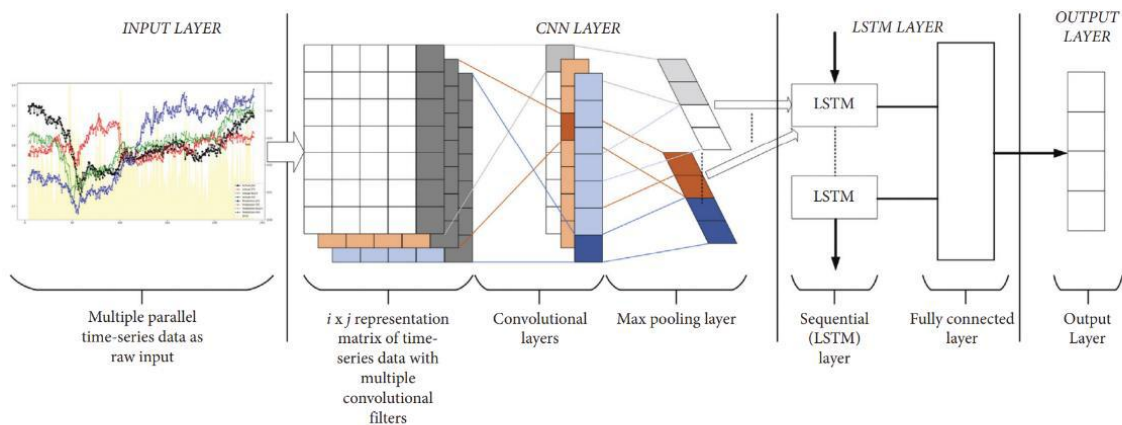


FIGURE 3: Architecture of the proposed multivariate CNN-LSTM for multiple parallel time-series prediction.

Why are we choosing 'x' model: We are trying to infer the amount of CO₂ sequestered by the Seaweed over time. Due to the temporal dimension, using an RNN (like LSTMs or GRUs) or a Transformer architecture is ideal for the Time-series model. Both models will use an MLP/feed-forward output layer

Why not other model(s): Using an MLP exclusively will hinder model performance as they aren't great at "understanding" temporal data (sequences) and images. The MLP struggles to understand dimensional aspects in the images from the Geospatial data and understand time-dependency from the Time-series data.

Tools: mentioned below are some models, tools, and libraries that could be integrated into the ai/ml model for more features.

Allometric Models: Allometric models provide a simple yet effective method for estimating carbon sequestration based on seaweed biomass. These models use the following formula:

$$C = k * B$$

Where:

C = carbon sequestered

k = species-specific conversion factor

B = biomass

The species-specific conversion factor (k) can be derived from research studies that have measured the carbon content of different seaweed species, such as Sargassum and Kappaphycus.

Data pipelines - GDAL (geospatial data), PyTorch (transforms and other).

Model creation - PyTorch (for both Time-series and Geospatial). Using PyTorch for Transfer learning too (if we use transfer-learning)

TRAINING:

Loss function: RMSE: Measurement of the level of accuracy can be carried out by applying the calculation of the root mean square error (RMSE) value to see the amount of deviation between the actual value and the resulting predictive value. . (11)

Data: the data is split into two parts: a training set and a test set.

Hyperparameter tuning: Parameters such as learning rate, batch size, number of neurons can be changed and optimized during training to find the best combination of parameters to reduce error

Validation: Validate the predicted output values of the test set to the actual values.

4.EXPECTED OUTCOMES:

4.1 Environmental Benefits:

Reduced CO₂ Levels: Seaweed cultivation is a promising method for carbon dioxide removal (CDR), leveraging high photosynthesis rates to absorb CO₂. Large-scale seaweed farming has been shown to effectively sequester carbon by trapping it in biomass, some of which sinks and gets buried in sediments 【12】 【13】 .

Predictive models can further optimize this process, enabling targeted carbon sequestration and enhancing ecosystem-based solutions 【12】 【13】 .

Improved Soil Quality: Seaweed-based bio-stimulants significantly enhance soil fertility, replacing chemical fertilizers with low-emission, sustainable alternatives. Their use has shown measurable improvements in nutrient cycling and pollutant degradation 【13】 .

Healthier Oceans: By reducing CO₂ in marine environments, seaweed farming mitigates ocean acidification, directly benefiting marine biodiversity and ecosystems 【12】 【13】 .

4.2 Agricultural and Ecosystem Benefits:

Dynamic Climate Adaptation:

AI-guided systems that adapt to changes in temperature and salinity have been suggested to stabilize ecosystems while increasing carbon. Seaweed farms restore marine habitats, fostering biodiversity by creating nutrient-rich environments for marine organisms.

Moreover, this approach supports broader ecosystem services, such as coral reef protection 【13】

4.3 Economic and Social Benefits:

Economic Growth: The market potential for seaweed-derived products like bioplastics and biostimulants highlights economic benefits. Seaweed offers a sustainable alternative to carbon-intensive inputs and plastics, presenting opportunities for significant revenue generation 【13】

Cost Savings: By reducing reliance on chemical inputs and enabling sustainable farming, seaweed-based solutions decrease costs associated with soil degradation and irrigation 【13】 .The proposal aligns with global sustainability strategies, but additional infrastructure and

data collection are necessary to maximize its potential. For further reading, explore detailed reports like those from Stanford University and The Nature Conservancy [12] [13] .

5. CHALLENGES AND LIMITATIONS

5.1 Challenges

5.1.1. Scalability-

Since there exists a lack of data available, this model can't be applied to a large biomass. The efficiency of carbon dioxide absorption by seaweed at larger scales may not be linear. Variables such as competition for nutrients, shading effects, and changes in water flow can reduce the predictability based on just the available data. We highly suggest the cultivation of new data sets. We further suggest adopting the following to derive optimal prediction rates. These two will continue training the model when new data sets enter the database.

- **Cloud Computing Infrastructure Solution:** Leverage cloud computing platforms (e.g., AWS, Google Cloud, Microsoft Azure) that store, process, and analyze large-scale datasets provide elastic computing resources, allowing the system to scale up or down depending on the workload. They also support distributed computing, enabling faster processing of complex models.
- **Data Compression and Sampling Solution:** Using data compression techniques or representative sampling to reduce the size of the dataset being processed while retaining key features reduces the amount of data that needs to be stored and processed, making scalability more achievable without significant computational cost.

5.1.2. Real-time Adaptation-

Real-time adaptation refers to the ability of the harvesting system to respond dynamically to changing environmental, operational, and ecological conditions. This is particularly challenging in seaweed biomass harvesting majorly due to the unpredictable ocean conditions. Factors like water temperature, salinity, currents, storms, and nutrient availability can change rapidly, making it difficult to maintain optimal growing and harvesting conditions. Here are a few methods in which this issue can be tackled.

- **Integration of Real-Time Satellite Data Solution:** Incorporate satellite-based remote sensing data to monitor sea surface temperature, chlorophyll levels (indicative of seaweed growth), and other oceanographic variables by using APIs from satellite data providers like NASA's MODIS or Copernicus Marine Service for continuous updates.
- **Continuous Monitoring with IoT and Sensors Solution:** Deploy IoT-enabled sensors in marine environments to collect real-time data on critical environmental factors such as water temperature, salinity, light intensity, and nutrient levels by using

low-cost, solar-powered sensors integrated into a network that transmits data to the predictive model via cloud infrastructure.

5.1.3. Invasive Species-

Biodiversity Monitoring Systems: Using drones, underwater cameras, and computer vision to monitor biodiversity in seaweed farming areas regulates biodiversity assessments, ensuring that farming practices do not negatively impact seaweed. Note that computer vision models (e.g., CNNs) can automatically detect and classify invasive seaweed in monitored areas.

5.2 Limitations

5.2.1. Economic Viability-

Since we strongly suggest accumulating more data, it is to be noted that this requires a high amount of capital. The full biophysical and economic limits to farming seaweed at scales relevant to the global carbon budget are not yet well understood, according to DeAngelo, J., Saenz, and others, in their research paper ‘Economic and biophysical limits to seaweed farming for climate change mitigation’ (21). Furthermore, achieving the modeled gigaton-scale climate benefits would require a >30-fold increase in the current global area of seaweed farming.

5.2.2. Carbon Sequestration Efficacy-

It is found in J DeAngelo, S Davis, B Saenz, and more’s research paper ‘Mapping seaweed farming potential for CDR and biomass products’ (3) that in the case of seaweed carbon removal, the carbon removed by sinking seaweed does not remain sequestered permanently, since large-scale ocean circulation might cause much of the carbon to resurface after a few hundred years. It is also unclear what fraction of seaweed carbon is ultimately removed from the atmosphere given dynamics of air-sea fluxes and downstream effects on biological productivity. These potential inefficiencies and uncertainties need to be considered when assessing the viability of sinking seaweed as a scalable carbon removal strategy. Due to which, further research in the field is required to fully validate carbon dioxide removal through seaweed biomass.

-----End of paper-----

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